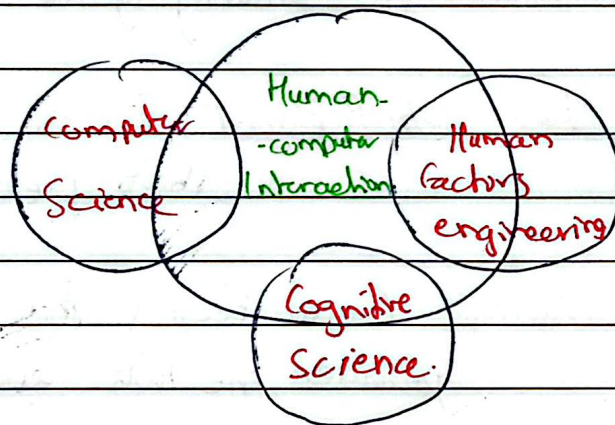


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Intro to HCI

Human Computer Interaction is a multi-disciplinary field of study, focusing on the design of computer technology.



Multidisciplinary field.

- It focuses on how humans interact with computational technologies & how to design, evaluate and implement interactive systems that safe & intuitive to use.

Goals of HCI:

The **ultimate goal** of HCI is to improve the relationship between humans & machines by optimizing system design around human capabilities, needs & limitations.

- 1) **Usability**: Ensuring systems are easy to learn.
- 2) **User Safety & Error Minimization**: Designing interfaces that prevent user errors.
- 3) **Utility**: Guaranteeing the software actually provides the tools and features users need.
- 4) **Positive UX**: Creating interfaces that are engaging, accessible & satisfying.
- 5) **Accessibility & Inclusivity**: Designing software so that all users, even disabled ones can use it.

Importance of HCI:

It can affect:

- effectiveness
- Productivity
- Morale
- Safety - designing

Variations in HCI

All users are different so designers must account for human variations across sensory, mental & social dimensions.

Human Factor	HCI	key design points
1) Human variations:	Differences in age, physical ability, culture & language	• provide multiple language options
2) Perceptual phenomenon	How humans sense & interpret visual, auditory & tactile input from screen	• Use strong color contrast, clear visual hierarchies & apply Gestalt Principles.

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3) Cognitive

How users process information, pay attention, reason & store memory.

Minimize cognitive load. e.g by avoiding overly complex menus.

4) Emotional.

The user's affective state (Joy, grief, anxiety)

Provide clear error messages & aesthetically layouts.

5) Motor

Physical Movement capabilities, fine motor control & hand-eye coordination.

Design larger click/tap targets for small screens or ageing users.

6) Social

How technology facilitates or alters communication & group dynamics.

Build clear indicators for presence & obvious privacy controls for sharing.

7) Behavioral / Psychological.

Habitual actions, interaction routines, user expectations.

Stick to familiar UI patterns so users don't have to learn basic navigation again

4 principles of scalable, user-friendly softwares:

1) Standardization

Adhering to established industry guidelines, rules & hardware/software norms across systems.

Benefit → Lowers the learning curve. Users rely on existing mental models instead of relearning how to navigate.

Example → Keyboard shortcuts like Ctrl+C & Ctrl+V ~~are considered~~ ~~are~~ have same copy/paste across multiple OS.

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2) Integration:

Connecting distinct software components, tools or services so they exchange data & work seamlessly.

Benefit → Saves user effort by eliminating repetitive data entry & switching friction between apps.

Example → Clicking a web link inside a messaging app to open it seamlessly in browser without interrupting your flow.

3) Consistency:

Keeping design elements, terminology, layouts & behaviours identical across the entire application or ecosystem.

Benefit → Prevents user confusion, reduces cognitive load & speeds up task execution.

Example → Using the exact same red color, icon style & modal prompt for every delete action across the app.

4) Portability:

The capability of a software or user interface to run effectively across different hardware platforms, OS versions or screen sizes.

Benefit → Ensures uniform access, flexibility & operational continuity regardless of device constraints.

example → Web apps like Figma, canva run identically on windows, macOS or linux.

AU 4:

• Consistency vs Standardization:

Internal

making app look & behave same across all its pages.

external

making app conform to universal industry rules.

• Integration + portability:

portable apps mostly rely on integrated ~~at~~ cloud backend APIs so users can start a task on a phone & finish it seamlessly on a desktop.

• Example (general):

Norman's door:

Don Norman said that if a door has a handle it should be a pull & with handle should be a push. This saves us from being confused & makes a door easy to use.

Usability measures:

How can we measure goodness of an ~~at~~ interface?

A few metrics have been defined to evaluate an interface:

Some examples are:

- 1) Task Success Rate (Effectiveness)
- 2) Time on Task (Efficiency)
- 3) Error Rate
- 4) System Usability Scale
- 5) Single ease Question
- 6) User feedback

Official standard for usability (ISO-9241).

This organizes all metrics into 3 pillars:

- 1) Effectiveness
- 2) Efficiency
- 3) Satisfaction